For graph G, WL iteration t, and colour label λ

$$\phi_t(G)[\lambda] = \sum_{v \mid c^{(t)}v = \lambda} \deg(v)$$

At WL iteration t, num of colours C_t is finite

so ~~so~~ we get finite dimensional vector

$$\phi_t(G) \in \mathbb{R}^{|C_t|}$$

So WL will stabilize after finite num of iterations ~~as num~~

$\therefore$ $K_{dWL}(G, H)$ is a finite sum

Hence, kernel is <u>well defined</u>

Kernel is a sum of dot product of vectors. Dot product is always positive semidefinite (PSD)

A sum of PSD kernels is also PSD.

Graphs $G_1, \ldots, G_n$ & scalars $m_1, \ldots, m_n \in \mathbb{R}$

$$\sum_{i,j} m_i m_j K_{dWL}(G_i, G_j) = \left\| \sum_i m_i \phi(G_i) \right\|^2 \geq 0$$

Therefore <u>kernel is also PSD</u>

It is also <u>symmetric</u>

Pseudocode

Initialize $c^0(v)$ $\forall v \in V$

We compute degree $\deg(V)$ for $\forall v \in V$

for $t = 0$ to T:

Initialize empty map ϕ_t

for each vertex $v \in V$:

$\lambda = c^t(v)$

$\phi_t[\lambda] = \phi_t[\lambda] + \deg(v)$

if $t < T$:

for each vertex $v \in V$:

$M =$ Multiset of colours $\{c^t(u) : u \in N(v)\}$

$c^{t+1}(v) = \text{Hash}(c^{(t)}(v), M)$

return $(\phi_0, \phi_1, \dots \phi_T)$

Then, compute kernel for graphs G, H:

$$K_{WL}(G, H) = \sum_t \langle \phi_t(G), \phi_t(H) \rangle$$

$n =$ num of vertices

$m = |E|$ edges

$T =$ WL iterations & processing neighbour multisets is $O(m)$

So complexity $= O(Tn + Tm) = O(T(m+n))$